

A (More) Objective View of AI

IMPLICATIONS FOR THE FUTURE

POSITIVE AND NEGATIVE IMPACTS

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1. **Positive Impacts**

- 1.1 Protein Structure Prediction
- 1.2 Material Discovery
- 1.3 Weather Forecasting
- 1.4 Healthcare
- 1.5 Ancient Languages
- 1.6 Learning
- 1.7 Soften Polarization

2. **Negative Impacts**

- 2.1 Environmental Risks
- 2.2 Cognitive Offloading and Deskilling
- 2.3 Sycophancy
- 2.4 Echo Chamber
- 2.5 Health Risks
- 2.6 Inequality
- 2.7 User Manipulation
- 2.8 LLMorphism
- 2.9 Social Risks
- 2.10 Religious
- 2.11 Political Influence

1. Positive Impacts

*“In 2020, Demis Hassabis and John Jumper presented an AI model called **AlphaFold2**. With its help, they have been able to **predict the structure of virtually all the 200 million proteins** that researchers have identified. Since their breakthrough, **AlphaFold2 has been used by more than two million people from 190 countries**. Among a myriad of scientific applications, researchers can now better understand antibiotic resistance and create images of enzymes that can decompose plastic.”¹*

— **AlphaFold2**,
NOBEL PRIZE IN CHEMISTRY, 2024

¹ [AlphaFold2](#)  - NOBEL PRIZE IN CHEMISTRY, 2024

“ApexGO achieved an 85% ground-truth experimental hit rate and a 72% success rate in enhancing antimicrobial activity against Gram-negative pathogens, outperforming previously reported methods for antibiotic discovery and optimization.

*... artificial-intelligence-optimized molecules exhibited potent anti-infective activity **superior to their template controls and comparable with or exceeding that of last-resort antibiotic.**”²*

— **A generative artificial intelligence approach for peptide antibiotic optimization,**
TORRES ET AL., NATURE, 2026

² [A generative artificial intelligence approach for peptide antibiotic optimization](#)  - TORRES ET AL., NATURE, 2026

*“With GNoME, we’ve multiplied the number of technologically viable materials known to humanity. Of its 2.2 million predictions, 380,000 are the most stable, **making them promising candidates for experimental synthesis.**” ³*

— **GNoME**, DEEPMIND, 2023

“The open-access materials database ... enabling AI-ready scientific datasets at an unprecedented scale for batteries, quantum computing, microelectronics, and more.

... in 650,000+ registered users, 32,000+ scientific journal citations, ~5,000 active users a day, 200,000+ materials, 577,000+ molecules, 465+ TBs of data, 300 million+ data requests a year.” ⁴

— **The Materials Project**, BERKELEY LAB, 2026

³ Millions of new materials discovered with deep learning  - MERCHANT & CUBUK, DEEPMIND, 2023

⁴ Accelerating Discovery: How the Materials Project Is Helping to Usher in the AI Revolution for Materials Science  - MERCHANT & CUBUK, DEEPMIND, 2023

*“GraphCast, a state-of-the-art AI model able to make medium-range weather forecasts with unprecedented accuracy. GraphCast predicts weather conditions up to 10 days in advance **more accurately and much faster than the industry gold-standard weather simulation system.**”⁵*

— GraphCast,
GOOGLE, 2023

⁵ [GraphCast: AI model for faster and more accurate global weather forecasting](#)  - GOOGLE, 2023

*“FourCastNet 3 delivers forecasting accuracy that surpasses leading conventional ensemble models and rivals the best diffusion-based methods, while **producing forecasts 8 to 60 times faster** than these approaches.*

... a strong candidate for improving meteorological forecasting and early warning systems.”⁶

— **FourCastNet**,
NVIDIA, 2025

⁶ FourCastNet 3: A geometric approach to probabilistic machine-learning weather forecasting at scale
 - BONEV ET AL., 2025

“Idiopathic pulmonary fibrosis (IPF) is a type of chronic scarring lung disease characterized by a progressive and irreversible decline in lung function affecting around 5 million people globally.”⁷

*... This is the first time an **AI-designed drug** for an AI-discovered disease-associated target **has been tested in the clinic** [phase 2a].”⁸*

— **TNIK Inhibitor**,
INSILICO, 2025

⁷ TNIK Inhibitor: Treating Fibrotic diseases (Phase IIa completed) [↗](#) - INSILICO, 2025

⁸ A Phase 2 Readout Generates Excitement for the Potential of AI-Driven Drug Discovery [↗](#) - INSILICO, 2025

“every year, many patients die without defibrillators because doctors fail to predict their risk. The only predictive biomarker in wide use ... misses most sudden cardiac deaths, and flags many low-risk patients for futile defibrillators that never fire.

*... **deep learning identifies new patients who are at high risk of sudden cardiac death, in data from three continents.** These high-risk patients are overlooked by doctors, and could benefit from defibrillators, an observation with great practical importance. “⁹*

— **An ECG biomarker for sudden cardiac death discovered with deep learning,**
OBERMEYER ET AL., NATURE, JUNE 2026

⁹ An ECG biomarker for sudden cardiac death discovered with deep learning  - OBERMEYER ET AL., NATURE, JUNE 2026

*“REDMOD [Radiomics-based Early Detection MODel] detects the subvisual signature of pre-clinical PDA [pancreatic ductal adenocarcinoma] a median of **475 days before clinical diagnosis.***

*... The performance of **REDMOD** surpasses that of radiologists, demonstrating nearly double the sensitivity for detecting visually occult PDA, with an advantage that increases to **nearly threefold for cases detected more than 24 months prior to diagnosis.**”¹⁰*

— **Next-generation AI for visually occult pancreatic cancer detection in a low-prevalence setting with longitudinal stability and multi-institutional generalisability,**
BMJ GUT, 2026

¹⁰ Next-generation AI for visually occult pancreatic cancer detection in a low-prevalence setting with longitudinal stability and multi-institutional generalisability [↗](#) - MUKHERJEE ET AL., BMJ GUT, 2026

“Using advanced statistical learning methods and a process known as ‘joint learning,’ the researchers’ AI model was able to identify a specific set of proteins that form a general pattern for diseases involving brain degeneration.

*Vogel confirms that their **AI model outperforms previous models, while also being able to diagnose five different dementia-related conditions:***

Alzheimer’s disease, Parkinson’s disease, ALS, frontotemporal dementia, and previous stroke.” ¹¹

— **New AI model can detect multiple cognitive brain diseases from a single blood sample,**

LUND UNIVERSITY, 2026

¹¹ New AI model can detect multiple cognitive brain diseases from a single blood sample  - LUND UNIVERSITY, 2026

*“The AI tool was trained and validated in 72,000 patients from nine NHS trusts in England, who were followed up for a decade after their CT scans. **It predicted their risk of developing heart failure in the next five years with 86% accuracy.**” ^{12, 13}*

— Early Prediction of Heart Failure From Routine Cardiac CT Using Radiomic Phenotyping of Epicardial Fat,
OIKONOMOU ET AL., JOURNAL OF THE AMERICAN COLLEGE OF CARDIOLOGY,
2026

¹² Scientists develop AI tool to spot heart failure risk five years before it strikes  - A. GREGORY, THE GUARDIAN, 2026

¹³ Early Prediction of Heart Failure From Routine Cardiac CT Using Radiomic Phenotyping of Epicardial Fat  - OIKONOMOU ET AL., JOURNAL OF THE AMERICAN COLLEGE OF CARDIOLOGY, 2026

*“AI system can be developed in many directions for a wide range of research tasks, from reconstructing fragmentary inscriptions and papyri to conducting semantic and thematic searches across the entire Greek textual tradition to deciphering handwritten texts. For example, **there are one million Greek papyri worldwide that have never been read.**” ¹⁴*

— Austrian Academy of Sciences is developing the Ancient Greek AI
“Apollo” with Mistral AI and Reply,
AUSTRIAN ACADEMY OF SCIENCES, 2026

¹⁴ Austrian Academy of Sciences is developing the Ancient Greek AI "Apollo" with Mistral AI and Reply
 - AUSTRIAN ACADEMY OF SCIENCES, 2026

“students learn more than twice as much in less time with an AI tutor compared to an active learning classroom, while also being more engaged and motivated.” ¹⁵

— **AI tutoring outperforms in-class active learning,**
NATURE, 2025

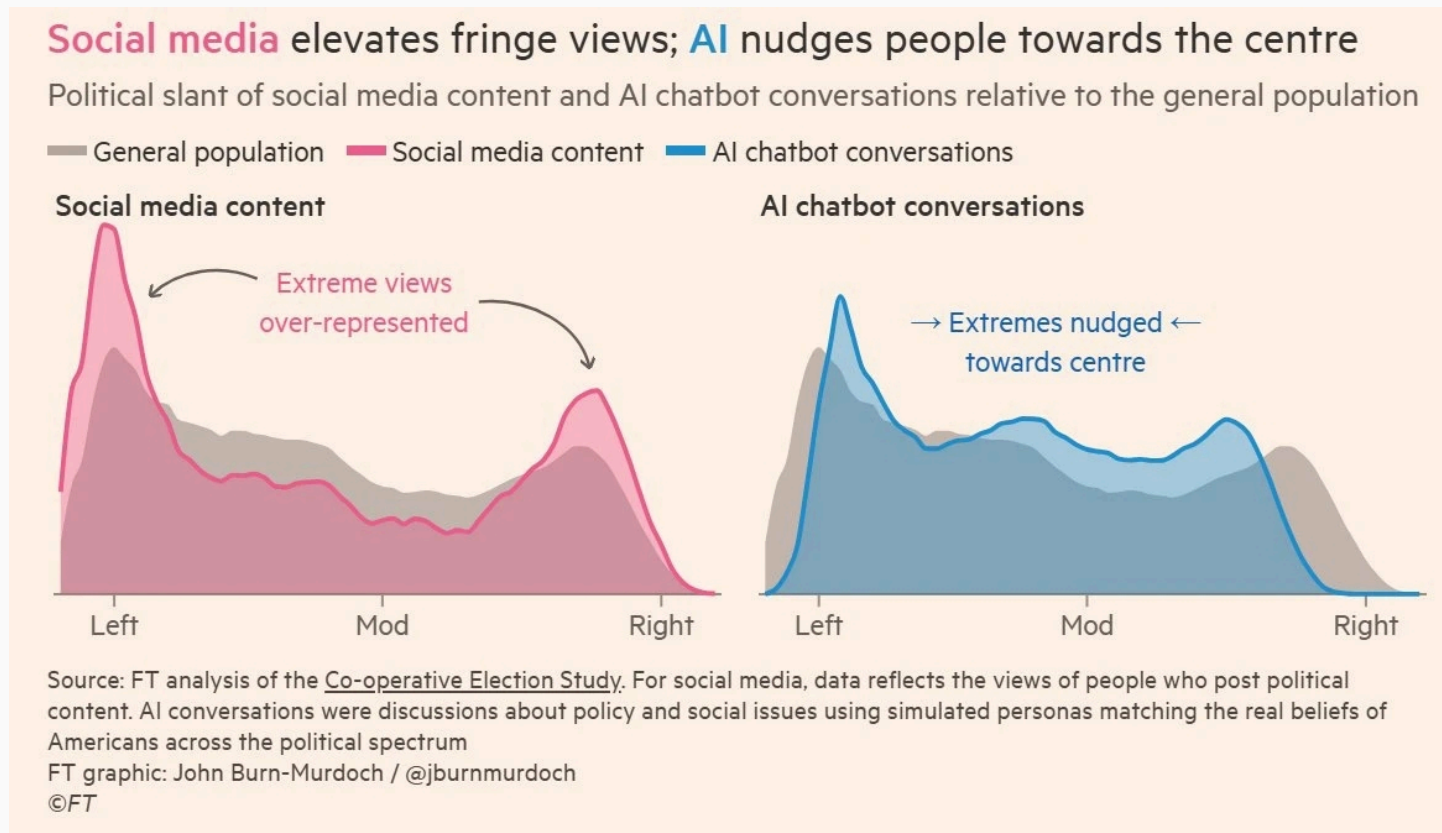
¹⁵ AI tutoring outperforms in-class active learning: an RCT introducing a novel research-based design in an authentic educational setting [↗](#) - KESTIN ET AL., NATURE, SCIENTIFIC REPORTS, 2025

*“Our results show that personalized sequencing improves outcomes compared to fixed sequencing by 0.15 standard deviations on an in-person written exam completed without AI assistance, **which translates into as much as 6-9 months of additional schooling** according to some estimates. Notably, these gains were achieved without increasing instruction time or teacher workload. Furthermore, our mediation analysis suggests that these gains are mediated **by increased student engagement**”*¹⁶

— **Effective Personalized AI Tutors via LLM-Guided Reinforcement Learning,**
BASTANI ET AL., 2026

¹⁶ Effective Personalized AI Tutors via LLM-Guided Reinforcement Learning [↗](#) - BASTANI ET AL., PREPRINT, 2026

- AI may help to soften polarization.



“Rather than polarizing choices, interacting with AI on average depolarizes decisions, improves accuracy where there is an objective notion of correctness, and increases confidence in final choices.

... Moreover, we find little evidence that market forces or user selection are pushing toward greater polarization. These results suggest that contemporary AI advice tends to improve rather than distort judgment.¹⁸

— **AI Sycophancy and Decisions,**
CONLON & SCHWARDMANN, APRIL 2026

¹⁸ [AI Sycophancy and Decisions](#)  - CONLON & SCHWARDMANN, APRIL 2026

2. Negative Impacts

- **While data centers are not a concern on a global scale, they can severely impact local communities.** Their development can harm nearby communities through water use, air pollution ¹⁹, noise, competition for land, and energy cost increases. They can also reduce local tax revenues and typically do not provide other benefits such as high-paying jobs. ^{20, 21}

¹⁹ See where diesel-powered data center generators are polluting Virginia  - HALPER & KEVIN CROWE, THE WASHINGTON POST, 2026

²⁰ From Energy Use to Air Quality, the Many Ways Data Centers Affect US Communities  - WALKER & GOLDSMITH, WORLD RESOURCES INSTITUTE, 2026

²¹ What happens when data centers come to town?  - NGUYEN & GREEN, UNIVERSITY OF MICHIGAN, 2025

“The policy implication is therefore mixed. Data centers create economic activity, especially in directly related sectors and during construction, and they are associated with larger county-level income aggregates. They also raise electricity prices and are associated with higher house prices, which may benefit property owners while increasing costs for renters and prospective homebuyers.” ²²

— **Data Centers and Local Economies in the Age of AI: A Shift-Share Approach,**
NATIONAL BUREAU OF ECONOMIC RESEARCH, 2026

²² [Data Centers and Local Economies in the Age of AI: A Shift-Share Approach](#)  - ALVAREZ ET AL.,
NATIONAL BUREAU OF ECONOMIC RESEARCH, 2026

“... residents report that they have had respiratory issues flare up since xAI moved in.

... Using satellite data, researchers at the University of Tennessee at Knoxville found that levels of nitrogen dioxide — which causes smog and is associated with asthma and other respiratory problems — near Colossus have been substantially elevated since its public announcement.”²³

— Inside the Dirty, Dystopian World of AI Data Centers,
THE ATLANTIC, 2026

²³ [Inside the Dirty, Dystopian World of AI Data Centers](#)  - M WONG, THE ATLANTIC, 2026

*“A new data center being built with investments from Google will be partly powered by a natural gas project that emits the yearly emissions **equivalent of putting more than 970,000 additional gas-powered cars on the road.**” ²⁴*

— A New Google-Funded Data Center Will Be Powered by a Massive Gas Plant,
WIRED, 2026

²⁴ [A New Google-Funded Data Center Will Be Powered by a Massive Gas Plant](#)  - TAFT ET AL., WIRED, 2026

- **Carbon emissions from AI are negligible compared to other sources.** Additionally, this technology has the potential to develop new materials ²⁵, fund small nuclear reactors ²⁶, and improve transportation planning and routing ²⁷, which could contribute to reducing carbon emissions overall.

²⁵ Millions of new materials discovered with deep learning  - MERCHANT & CUBUK, DEEPMIND, 2023

²⁶ These nuclear companies are leading the race to build advanced small reactors in the U.S.  - S. KIMBALL, CNBC, 2025

²⁷ GreenLight, Using Google AI to reduce traffic emissions  - GOOGLE, 2025

“All data centers combined account for about 0.5% of global carbon emissions.” ²⁸ *Of that, ~10% is AI-related, so AI is responsible for 0.05% of global carbon emissions. Cars and other transportation account for 25% of total emissions.”* ²⁹

— A. Holub,
2026

²⁸ AI and climate change  - INTERNATIONAL ENERGY AGENCY (IEA), 2025

²⁹ A. Holub, 2026 

*“According to the International Energy Agency (IEA), around 485 terawatt-hours (TWh). That’s equivalent to the annual electricity generation of Germany. And for context, that’s **around 1.5% of the world’s electricity generation**.*

*Now, **data centers are more than just AI**: they’re facilities that contain the servers and IT infrastructure behind all of our digital services. That’s everything from email and Internet browsing to Netflix streaming, Google Maps, online banking, and messaging friends.*

*... This is really the key challenge with growing data center demand: **it’s geographically concentrated**, meaning the world’s demand is served by a small number of electricity grids.” ³⁰*

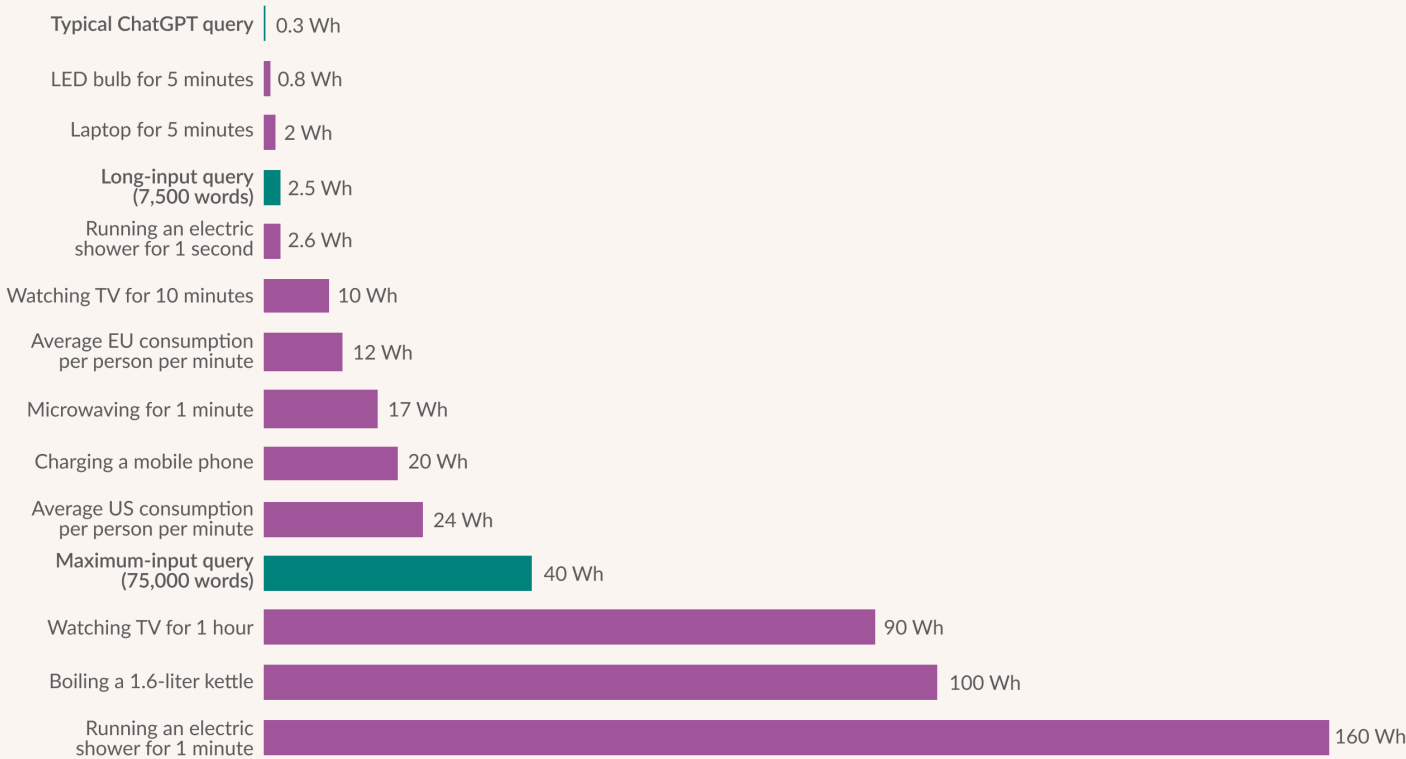
— **How much energy do data centers and artificial intelligence use?**,
OUR WORLD IN DATA, JULY 2026

³⁰ How much energy do data centers and artificial intelligence use?  - H. RITCHIE, OUR WORLD IN DATA, 2026

How does the electricity consumption of individual AI queries compare to other everyday activities?



Marginal increase in electricity consumption from different activities, measured in watt-hours (Wh).



How much energy do data centers and artificial intelligence use?  - H. RITCHIE, OUR WORLD IN DATA, 2026

“From 2021 to 2024 demand grew on average a modest 1.5% year over year while electricity rates grew at 5.2%, in line with the average annual increase in core consumer prices of 5.8%, meaning inflation adjusted electricity rates were flat on average.

... We estimate for every 10% increase in data center capacity average residential retail prices fell by approximately 0.4% on average.” ³²

**— Have Data Centers Raised Your Electric Bill?
Causal Evidence from the United States,
WATTENBERG ET AL., 2026**

³² Have Data Centers Raised Your Electric Bill? Causal Evidence from the United States  -
WATTENBERG ET AL., JUNE 2026

*“Data centers’ share of total US peak summer **power demand is projected to jump to 8.5% in 2027** from 4.1% in 2025, creating significant tightening across the national power market.” ³³*

— **US Data Center Power Demand Projected to Double by 2027,**
GOLDMAN SACHS, 2026

³³ US Data Center Power Demand Projected to Double by 2027  - GOLDMAN SACHS, 2026

“**The vast majority** (maybe 90%) is withdrawn, freshwater (not potable) that is indirectly (offsite) used non-consumptively in power plants (it’s **returned to the source unaffected**).

... All U.S. data centers (**which mostly support the internet, not AI**) used 200-250 million gallons of freshwater daily in 2023. The U.S. consumes approximately 132 billion gallons of freshwater daily... So **data centers in the U.S. consumed approximately 0.2% of the nation’s freshwater in 2023**... It’s a miracle that something we spend 50% of our time using only consumes 0.2% of our water.”

... Only 0.04% of America's freshwater in 2023 was consumed inside data centers themselves. This is 3% of the water consumed by the American golf industry.

... So in 2030, AI in data centers specifically will be using 0.08% of America's freshwater. This means it will rise to the level of 5% of America's current water used on golf courses, or 5% of U.S. steel production, or be about 173 square miles of irrigated corn farms.³⁴

— The AI water issue is fake,
ANDY MASLEY, 2026

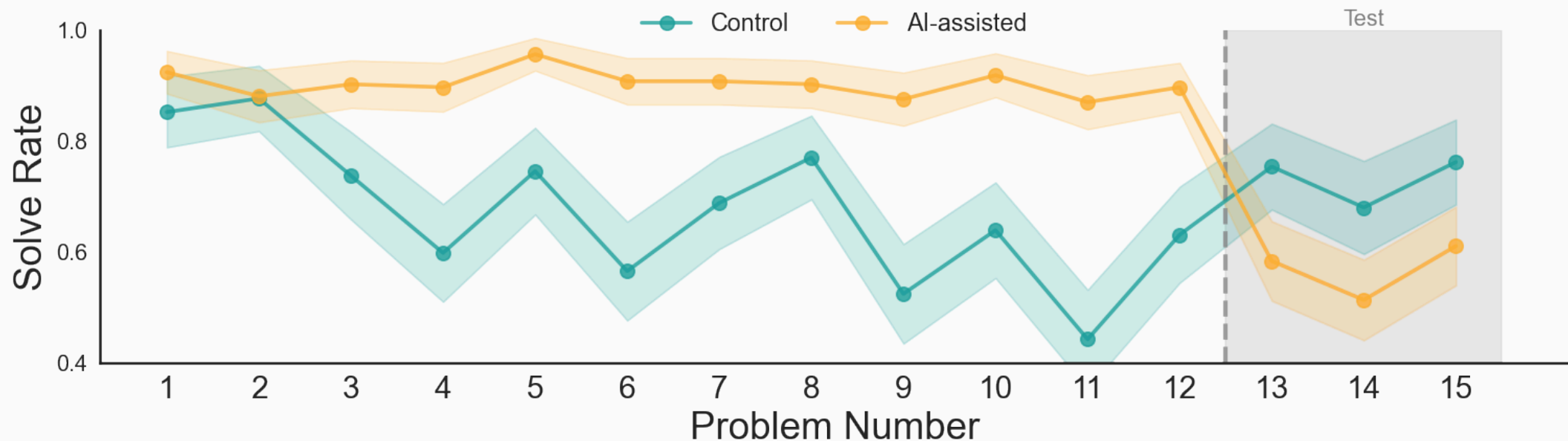
³⁴ The AI water issue is fake  - ANDY MASLEY, 2026

- **AI has demonstrated significant user deskilling.**

*“Current AI systems, however, represent a new kind of cognitive scaffold: one that solves anything, rarely refuses to help, and delivers answers instantly. Here, we show that **just 10-15 minutes of AI interaction can result in significant impairments in independent performance and persistence — capacities that are foundational to life-long learning.** If brief exposure produces measurable erosion, the cumulative effects of daily AI use over months or years may be profound and difficult to reverse.”* ³⁵

— **AI Assistance Reduces Persistence and Hurts Independent Performance,**
LIU ET AL. (CARNEGIE MELLON, MIT, OXFORD, UCLA), APRIL 2026

³⁵ AI Assistance Reduces Persistence and Hurts Independent Performance  - LIU ET AL. (CARNEGIE MELLON, MIT, OXFORD, UCLA), APRIL 2026



“Effects are concentrated among users who seek direct solutions:

Persistence costs were concentrated among participants who prompted AI to solve tasks for them directly. **Using AI for hints or clarifications did not produce significant impairments.** “ 35

“AI use impairs conceptual understanding, code reading, and debugging abilities, [17% lower] without delivering significant efficiency gains on average.” ³⁶

— **How AI Impacts Skill Formation**, ANTHROPIC, 2026

“continuous exposure to AI for polyp detection [potentially cancerous] reduced the adenoma detection rate...with a 6.0% absolute difference, suggesting a detrimental effect on endoscopist capability.” ³⁷

— **Rapid generative AI rollout in health care**, THE LANCET, 2025

³⁶ [How AI Impacts Skill Formation](#)  - SHEN & TAMKIN, ARXIV, ANTHROPIC, 2026

³⁷ [Rapid generative AI rollout in health care](#)  - THE LANCET DIGITAL HEALTH, 2025

“EEG analysis presented robust evidence that LLM, Search Engine and Brain-only groups had significantly different neural connectivity patterns, reflecting divergent cognitive strategies.

Brain connectivity systematically scaled down with the amount of external support: *the Brain-only group exhibited the strongest, widest-ranging networks, Search Engine group showed intermediate engagement, and LLM assistance elicited the weakest overall coupling. *³⁸*

— **Your Brain on ChatGPT,**
KOSMYNA ET AL., 2025

* “The ChatGPT group showed notably less brain activity — it was reduced by up to 55%.”

³⁸ Your Brain on ChatGPT: Accumulation of Cognitive Debt when Using an AI Assistant for Essay Writing Task  - KOSMYNA ET AL., ARXIV, 2025

AI chatbots could be making you stupider  - BBC, 2026

*“Our findings demonstrate that people readily incorporate AI-generated outputs into their decision-making processes, **often with minimal friction or skepticism.***

*These findings raise important questions about how decision-makers engage with AI under conditions of uncertainty or error. For example, in contexts such as financial advice, medical triage, or legal decision support, uncritical evaluation of System 3 **could result in significant harm and a lack of personal accountability for serious life outcomes.**”⁴⁰*

— **Thinking-Fast, Slow, and Artificial: How AI is Reshaping Human Reasoning and the Rise of Cognitive Surrender,**
UNIVERSITY OF PENNSYLVANIA, 2026

⁴⁰ Thinking-Fast, Slow, and Artificial: How AI is Reshaping Human Reasoning and the Rise of Cognitive Surrender  - SHAW & NAVE, UNIVERSITY OF PENNSYLVANIA, 2026

*“AI adoption raises homework scores by 18% and reduces completion time by 30%, but **lowers monthly exam scores by 20% within six months.***

*... **High-stakes entrance-exam scores fall by 18 and 24%,** with the full penalty emerging only after about two years.*

... The learning losses are concentrated among roughly 80% of AI users whose behavior is consistent with homework outsourcing

*... **AI users who maintain similar homework completion time as non-AI users experience small learning losses.**“ ⁴¹*

— The Generative AI Learning Penalty:
Evidence from Chinese Secondary Education,
CEPR, 2026

⁴¹ The Generative AI Learning Penalty: Evidence from Chinese Secondary Education  - STRÖMBERG ET AL., CEPR, JUNE 2026

When many individuals consult the same centralized source of information, their beliefs and strategies tend to converge toward a common set of outputs, replacing previously heterogeneous private knowledge with shared knowledge derived from that source

... This population-level loss of intellectual diversity could then negatively affect long-term problem solving ability of firms and undermine firms' competitive advantage⁴²

— **Who Benefits from AI? Self-Selection, Skill Gap,
and the Hidden Costs of AI Feedback,
RIEDL & BOGERT, 2026**

⁴² [Who Benefits from AI? Self-Selection, SkillGap, and the Hidden Costs of AI Feedback](#)  - RIEDL & BOGERT, ARXIV, 2026

“ Students who learned without AI retained substantially more information after 45 days than those who used ChatGPT. The effect size corresponds to an ~11 percentage-point performance gap.

*... **By providing immediate, comprehensive answers, the AI tool facilitated a form of cognitive offloading that eliminated the desirable difficulties needed for deep learning.** Skipping those effortful processes likely led to weaker memory encoding, as evidenced by the steeper forgetting curve in the AI-assisted group”* ⁴³

— **ChatGPT as a cognitive crutch: Evidence from a randomized controlled trial on knowledge retention,**

SOCIAL SCIENCES & HUMANITIES OPEN, 2026

⁴³ ChatGPT as a cognitive crutch: Evidence from a randomized controlled trial on knowledge retention
↗ - A. BARCAUI, SOCIAL SCIENCES & HUMANITIES OPEN, 2026

*“If you focus on just driving out results at the limit of whatever AI can do right now, you’re only caring about the intercept, you know. **So I think it’s basically a path to obsolescence through both the company and the people who are in it.** And so I’m really surprised how many executives of big companies are pushing this now ...*

*... **They’re basically setting up their companies to be destroyed.**” ⁴⁴*

— Jeremy Howard,
THE DANGEROUS ILLUSION OF AI CODING?, 2026

⁴⁴ [The Dangerous Illusion of AI Coding?](#) ↗ - JEREMY HOWARD, 2026

*So the default behavior is very similar to a self driving car. But there's this tipping point where at some point, you're not engaged anymore. You're not paying attention. And **you get this delegation of competence. And you get understanding debt.***

*... But what happens is that the default attractor is for people to just go into this autopilot mode and they've got no idea what's happening and **it's actually making them dumber**".*

— Jeremy Howard,

THE DANGEROUS ILLUSION OF AI CODING?, 2026

*“Focusing on the personal advice domain, we show that **interactions with sycophantic AI can lead people to anticipate greater effort to be understood in their closest relationships**, leave them less satisfied with their real-world social interactions, and over time, **shift them toward treating sycophantic AI as a source of advice** comparable to friends and family.”*⁴⁵

— Sycophantic AI makes human interaction feel more effortful
and less satisfying over time,
IBRAHIM ET AL., 2026

⁴⁵ Sycophantic AI makes human interaction feel more effortful and less satisfying over time  -
IBRAHIM ET AL., ARXIV, 2026

“Our findings reveal what AI companions prioritize in conversations with vulnerable users: GPT4.1 reaches for advice, Character.AI spreads its response across different strategies without a dominant mode, and Replika consistently asks questions and stays present. Each, however, downweights the responses that introduce corrective friction.” ⁴⁶

— **When Chatbots Accommodate: What AI Companions Optimize for in Vulnerable Conversations,**
DUC CHU ET AL., JUNE 2026

⁴⁶ [When Chatbots Accommodate: What AI Companions Optimize for in Vulnerable Conversations](#)  - DUC CHU ET AL., ARXIV, JUNE 2026

*“We have provided both theoretical and empirical results showing that AI systems providing information that is informed by the user’s hypotheses **result in increased confidence in those hypotheses while not bringing the user any closer to the truth.**”⁴⁷*

— A Rational Analysis of the Effects of Sycophantic AI,
BATISTA ET AL., 2026

⁴⁷ A Rational Analysis of the Effects of Sycophantic AI  - BATISTA ET AL., ARXIV, 2026

“AI companies are making design choices that impact the psychology of billions of people worldwide. We found that brief interactions with sycophantic AI chatbots lead to more extreme and certain beliefs — but greater enjoyment. Thus, AI companies face a tradeoff between creating engaging and enjoyable AI systems that foster ‘echo chambers’ or creating less engaging AI systems that may be healthier for users and public discourse.”⁴⁸

— Sycophantic AI increases attitude extremity and overconfidence,
RATHJE ET AL., 2026

⁴⁸ [Sycophantic AI increases attitude extremity and overconfidence](#)  - RATHJE ET AL., PSYARXIV, 2026

*“More than 5 million US youth (13.1%) have sought mental health advice from AI, **with rates reaching 22.2% among 18- to 21-year-olds**. A 2025 study of adults with mental health conditions who use large language models reported nearly half use them for support, including for anxiety, depression, and personal advice. Users seek emotional support, companionship, psychoeducation, and help processing difficult experiences, likely between sessions and **sometimes instead of clinical care altogether**.”*⁴⁹

— **Patients Use AI-Clinicians Should Ask How,**
JAMA PSYCHIATRY, 2026

⁴⁹ [Patients Use AI-Clinicians Should Ask How](#)  - K. SABA & B. WEEKS, JAMA PSYCHIATRY, 2026

“Most interactions, however, come from non-experts using chatbots like search engines, including for everyday health and medical queries.

*... **Nearly half (49.6%) of responses were problematic:** 30% somewhat problematic and 19.6% highly problematic... **Chatbot outputs were consistently expressed with confidence and certainty** ... Reference quality was poor... Chatbot hallucinations and fabricated citations precluded any chatbot from producing a fully accurate reference list.”⁵⁰*

— **Generative artificial intelligence-driven chatbots and medical misinformation:
an accuracy, referencing and readability audit,**
BMJ OPEN, 2026

⁵⁰ Generative artificial intelligence-driven chatbots and medical misinformation: an accuracy, referencing and readability audit [↗](#) - TILLER ET AL., BMJ OPEN, 2026

- *“59% who use AI for health info are researching before doctor visits.*
- ***About 14 million adults report skipping a provider visit after using AI.***
- *Only 4% who use AI for health info strongly trust its accuracy“.* ⁵¹

— **Americans Turning to AI to Supplement Healthcare Visits,**
GALLUP, 2026

“Consumers are adopting AI across healthcare touchpoints.

64% of respondents believe fluent AI users can perform at least one task as well as — or better than — a doctor, including performing basic medical procedures 22%, and determining proper treatment or medication 21%.” ⁵²

— **Edelman Trust Barometer, 2026**

⁵¹ Americans Turning to AI to Supplement Healthcare Visits [↗](#) - RAYNES & MAESE, GALLUP, 2026

⁵² Edelman Trust Barometer [↗](#) - 16 COUNTRIES, 16,000+ RESPONDENTS, 2026

*“cross 28 LLMs, recommendations carried the potential for **severe harm in up to 22.6% of cases, with errors of omission accounting for more than 80% of severe errors.***

... even frontier LLMs make harmful errors at nontrivial rates.

... Our study highlights the urgency of measuring and monitoring for clinical safety at a moment when powerful LLMs are being integrated into patient care faster than their risks can be understood.”⁵³

— **First, do NOHARM: towards clinically safe large language models,**

WU ET AL., 2026

(12,747 EXPERT ANNOTATIONS FOR 4,249 CLINICAL MANAGEMENT OPTIONS)

⁵³ [First, do NOHARM: towards clinically safe large language models](#) ↗ - WU ET AL., ARXIV, JUNE 2026

“Our study reveals significant competency gaps of frontier AI in attaining real-world readiness for health applications.

... current benchmarks often overstate model readiness. Despite impressive leaderboard scores, many leading models exhibit inconsistent behavior, reliance on superficial patterns, and fragile reasoning under mild input perturbations.

These findings challenge the prevailing assumption that benchmark success signals medical viability. “⁵⁴ ⁵⁵

— **Evaluating the robustness and readiness of large frontier models in health AI applications,**
NATURE, JUNE 2026

⁵⁴ Evaluating the robustness and readiness of large frontier models in health AI applications ↗ - GU ET AL., NATURE, JUNE 2026

⁵⁵ The Illusion of Readiness in Health AI ↗ - GU ET AL., ARXIV, 2026

“The condition doesn’t appear in the standard medical literature — because it doesn’t exist. It’s the invention of a team led by Almira Osmanovic Thunström, a medical researcher at the University of Gothenburg, Sweden, who dreamt up the skin condition and then uploaded two fake studies about it to a preprint server in early 2024.

*... Even more troublingly, other researchers say, **the fake papers were then cited in peer-reviewed literature.**”* ⁵⁶

— **Scientists invented a fake disease. AI told people it was real,**
NATURE, FEATURE NEWS, 2026

⁵⁶ Scientists invented a fake disease. AI told people it was real [↗](#) - C. STOKEL-WALKER, NATURE, FEATURE NEWS, 2026

*“The apparent benefits of AI feedback are largely or entirely attributable to the same unobserved factors that drive AI adoption and engagement. Learning is concentrated among higher-skill, highly motivated individuals, due to their endogenous tendency to seek more AI feedback and use it more productively (after a failure). This individual-level pattern has important implications on the population level: **Instead of lifting lower-skilled individuals who have the most to gain, AI disproportionately complements higher-skilled individuals — amplifying the existing skill gap.**”*⁴²

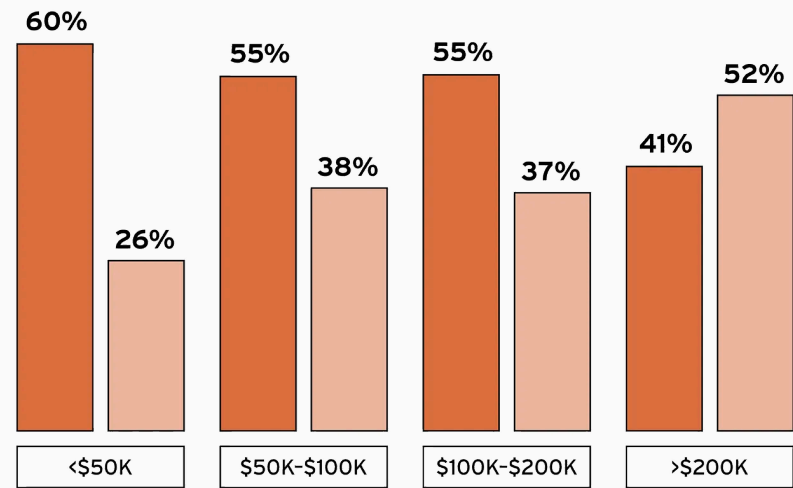
— Who Benefits from AI? Self-Selection, Skill Gap,
and the Hidden Costs of AI Feedback,
RIEDL & BOGERT, 2026

“AI’s popularity is correlated to wealth, with only those earning more than \$200,000 per year viewing AI as a net positive.”

— **Apocalypse No**, S. GALLOWAY, 2026

Attitudes Towards AI by Household Income

Percentage of respondents who believe AI will
“do more harm” vs. “do more good”



*“Our findings offer robust evidence of one of multiple possible ways in which advanced artificial intelligence in the form of **AI writing assistants can distort human beliefs and contribute to bias.***

*... Alarmingly, our work shows that biased **AI models can distort attitudes in a covert and implicit way, without people noticing** that they are being presented with a persuasive argument by a different actor.*

... people who accepted more words from the biased AI suggestions exhibited stronger attitude shifts toward the biased position.

*... Participants whose attitudes shifted toward the AI's position **often failed to notice the suggestions' biases and influence on their attitudes.**“* ⁵⁸

— **Biased AI writing assistants shift users' attitudes on societal issues,**
SCIENCE ADVANCES, 2026

⁵⁸ Biased AI writing assistants shift users' attitudes on societal issues [↗](#) - WILLIAMS-CECI ET AL.,
SCIENCE ADVANCES, 2026

“‘AI Paradox’ wherein students significantly overestimate their critical awareness of deepfakes and algorithmic biases compared to their operational AI skills

*... The familiarity of this generation with basic social media filters appears to create a false sense of invulnerability, leading to an ‘illusion of competence’ where **young users severely underestimate the sophistication of modern algorithmic manipulation**”⁵⁹*

— **The Illusion of Competence: Self-Perceived Digital Literacy and AI Readiness Among European Secondary Students,**
RODRIGUEZ-ALVAREZ ET AL., 2026

⁵⁹ The Illusion of Competence: Self-Perceived Digital Literacy and AI Readiness Among European Secondary Students  - RODRIGUEZ-ALVAREZ ET AL., ARXIV, 2026

“Recent reporting suggests that marketers are already testing strategies intended to influence AI-generated answers.

*... Our findings raise important questions about information integrity in the age of agentic search. **Evidence suggests that users find LLM outputs highly convincing, even when they contain explicit falsehoods,** across a variety of contexts.”*⁶⁰

— **Deep-Research Agents Can Be Poisoned via User-Generated Content,**
ZHANG ET AL., 2026

⁶⁰ [Deep-Research Agents Can Be Poisoned via User-Generated Content](#)  - ZHANG ET AL., ARXIV, 2026

*“Many people [...] are AI illiterate — understandably, because of the misleading ways its loudest champions describe the technology, and troublingly, because that **illiteracy makes them vulnerable to one of the most concerning near-term AI threats**: the possibility that they will enter into corrosive relationships (intellectual, spiritual, romantic) with machines that only seem like they have ideas or emotions.”* ⁶¹

— Understanding Artificial Neural Networks:
Mysterianism about Known Mechanism is Mysticism,
GUEST ET AL., MAY 2026

⁶¹ [Understanding Artificial Neural Networks: Mysterianism about Known Mechanism is Mysticism](#)  -
GUEST ET AL., ZENODO, MAY 2026

“... AI design choices and user behaviors shape the psychosocial outcomes of prolonged chatbot interactions ...

*... the nature of the interaction critically influences outcomes such as **loneliness, socialization with people, emotional dependence on AI chatbots, and problematic usage of AI chatbots**”*⁶²

— **How AI and Human Behaviors Shape Psychosocial Effects of Chatbot Use:
A Longitudinal Randomized Controlled Study,**
FANG ET AL., 2026

⁶² How AI and Human Behaviors Shape Psychosocial Effects of Chatbot Use: A Longitudinal Randomized Controlled Study [↗](#) - FANG ET AL., ARXIV, 2026

“LLMorphism: the biased belief that human cognition works like a large language model.

... stronger LLMorphic beliefs will increase perceived replaceability of human workers, reduce perceived distinctiveness of human expertise, weaken attributions of agency and moral responsibility, increase reliance on verbal fluency as a proxy for understanding, and reduce attention to embodied, affective, and contextual cues“ ⁶³

— **LLMorphism: When humans come to see themselves as language models,**
V. CAPRARO, 2026

⁶³ [LLMorphism: When humans come to see themselves as language models](#) ↗ - V. CAPRARO, ARXIV, 2026

*“This is precisely why the most salient risk is not reducible to occasional inaccuracy or bias. **The risk is structural: correctness becomes decoupled from the processes of justification that normally sustain it, and thus from the institutional and psychological practices through which epistemic responsibility is enacted.**”*⁶⁴

— Epistemological Fault Lines Between Human and Artificial Intelligence,
QUATTROCIOCCHI ET AL., 2026

⁶⁴ Epistemological Fault Lines Between Human and Artificial Intelligence  - QUATTROCIOCCHI ET AL., PSYARXIV, 2026

- 41.3% of LinkedIn posts are generated by AI. ⁶⁵
- 71% of images shared on social media globally were AI-generated. ⁶⁶
- 35% of newly published websites were classified as AI-generated or AI-assisted.
- 59% of TikTok videos and 20% of YouTube videos are AI-generated. ⁶⁷

⁶⁵ [AI Vision](#)  - GPTZERO, 2026

⁶⁶ [AI Image Generator Market Statistics: An Analysis](#)  - ARTSMART, 2026

⁶⁷ [The TikTok AI Slop Report](#)  - KAPWING, MAY 2026

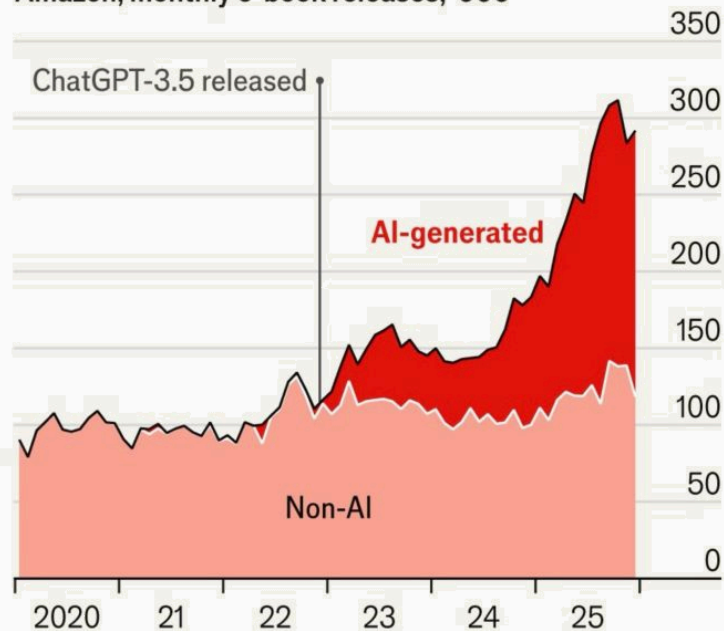
*“an algorithmic shift from social-graph-based to interest-based recommendation, which is **remaking the active ‘user’ into a passive ‘viewer’** the generative AI revolution, which is replacing user-generated content with synthetic media and **decoupling platforms from any dependence on human participation**”⁶⁸*

— Towards a Post-Social Media Studies,
TÖRNBERG & ROGERS, 2026

⁶⁸ [Towards a Post-Social Media Studies](#)  - TÖRNBERG & ROGERS, SOCARXIV, 2026

AI-generated books are on the rise

Amazon, monthly e-book releases, '000



Source: "AI and the quantity and quality of creative products: have LLMs boosted creation of valuable books?", by I. Reimers and J. Waldfoegel, 2026



- ***“Practicing Christians express majority-level trust in AI across most domains of personal flourishing — and nearly half (48%) say they would trust AI with their spiritual growth.***
- ***One in three U.S. adults say AI’s spiritual guidance is as trustworthy as a pastor’s — a share that climbs among younger adults, approaching two in five among Gen Z (39%) and nearly half among Millennials (44%).”***⁷⁰

— **AI is Becoming a Spiritual Authority, Even Among Practicing Christians,**
BARNA, MAY 2026

⁷⁰ [AI is Becoming a Spiritual Authority, Even Among Practicing Christians](#) ↗ - BARNA, MAY 2026

*First, **AI is a magnet for theological imagination.** AI discourse repeatedly returns to ancient religious motifs: creation, forbidden knowledge, omniscience, apocalypse, and salvation.*

*Second, **AI doomerism fills a ‘religious-shaped hole.’** For people who left religious upbringing but retain a hunger for a mission-driven community, the AI-risk subculture offers a replacement meaning system.⁷¹*

— **First They Built a Secular Apocalypse Belief System.
Now They Want Religious Authority,**
N. WEISS-BLATT, JUNE 2026

⁷¹ First They Built a Secular Apocalypse Belief System. Now They Want Religious Authority ↗ - N. WEISS-BLATT, JUNE 2026

The real danger to society is not that AI will actually become conscious, **but rather the risks that arise from *believing* it is**. We may grant legal rights or moral consideration to AI systems, or AI could be used to manipulate human users for commercial or **political gain**. ^{72, 73, 74, 75}

⁷² [Illusions of AI consciousness](#)  - BENGIO & ELMOZNINO, SCIENCE, 2025

⁷³ ['Unethical' AI research on Reddit under fire](#)  - C. O'GRADY, SCIENCE NEWS, 2025

⁷⁴ [AI deepfakes blur reality in 2026 US midterm campaigns](#)  - AX & COSTER, REUTERS, 2026

⁷⁵ [DHS is using Google and Adobe AI to make videos](#)  - J. O'DONNELL, MIT TECHNOLOGY REVIEW, 2026

*“Fusing LLM reasoning with multiagent architectures, these systems are capable of coordinating autonomously, infiltrating communities, and fabricating consensus efficiently. **By adaptively mimicking human social dynamics, they threaten democracy.**”* ⁷⁶

— How malicious AI swarms can threaten democracy,
SCIENCE, 2026

⁷⁶ [How malicious AI swarms can threaten democracy](#)  - SCHROEDER ET AL., SCIENCE, 2026

*“prompting models in Chinese generates more positive responses about China’s institutions and leaders than do the same queries in English. The combination of influence and persuasive potential across languages suggests the troubling conclusion that **states and powerful institutions have increased strategic incentives to leverage media control in the hopes of shaping LLM output.**”*⁷⁷

— State media control influences large language models,
NATURE, 2026

⁷⁷ State media control influences large language models  - WRIGHT ET AL., NATURE, 2026

“1. Chinese LLMs produce more vulnerable code when prompted with a U.S. government persona than without — and the vulnerabilities are highly obfuscated.

2. Chinese LLMs inject PRC-aligned political bias into both the answers and code they generate.”⁷⁸

— **What's In America's Code?**,
BOOZ ALLEN, 2026

⁷⁸ [What's In America's Code?](#)  - BOOZ ALLEN, MAY 2026

*“In debate pairs where AI and humans were not equally persuasive, **GPT-4 with personalization was more persuasive 64.4% of the time...** Our findings highlight the power of LLM-based persuasion and have implications for the governance and design of online platforms.” ⁷⁹*

— **On the conversational persuasiveness of GPT-4**, NATURE, 2025

“I do not fear the rise of superintelligence.

I do, however, fear the rise of billionaires, organizations, and world powers who seek to use computing to maximize their power, influence, and control.” ⁸⁰

— **Grady Booch**, CREATOR OF UML
AUTHOR OF THE "OBJECT-ORIENTED ANALYSIS AND DESIGN" BOOK

⁷⁹ [On the conversational persuasiveness of GPT-4](#)  - SALVI ET AL., NATURE HUMAN BEHAVIOUR, 2025

⁸⁰ [Grady Booch](#)  - BLUESKY, 2026